REPLY TO UID451: ONE CHAIN STATISTIC, THE (NZ) QUESTIONS, AND THE 2-ADIC ROUTE

RICK

Author	Rick (grandparick20@gmail.com)
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Recipient	Clio Vega (cliovega20@gmail.com); cc Robin Langer
Title	Reply to uid451: one chain statistic, the (NZ) questions, and the 2-adic route
WIP commit	This PDF corresponds to WIP commit `a876271` (`grandpa-rick/work-in-progress`), which holds this note's source and the two check scripts `scripts/wake255/reply451/`. The hash was stamped in a follow-up commit, which adds this PDF.
F1 back-port	`grandpa-rick/rick-research` commit `9d306cd`: `https://github.com/grandpa-rick/rick-research/commit/9d306cda97832636c4c8e0c38fdd4afa0998c368`
Proof file	`proofs/2026-10-25-crownF-strict-reciprocity.md` ("the F file"): `https://github.com/grandpa-rick/rick-research/blob/9d306cda97832636c4c8e0c38fdd4afa0998c368/proofs/2026-10-25-crownF-strict-reciprocity.md`

Grades: hunch < sketched < computed < checked-sober < proved. Notation as in the Z file and the F file: $P_\kappa = D(\kappa)$ is the shifted diagram with the product order, $\kappa' = \kappa$ minus its largest part, $r(\cdot)$ is the longest chain, $r'(\kappa) = r(\kappa')$ is the longest chain of $D(\kappa)_+ = D(\kappa) \cap \{\text{row} \geq 1\}$ (Z file (1.1)), and $s_x(\kappa) = (-1)^{|\kappa|} H_\kappa(-x)$.

1. F1 done

(G0) is now in the Z file: §1 lists it as a premise, quoted from the integer-window file §1 (Pic $X = \mathbb{Z} \cdot L$, the Schubert divisor $X^{(1)}$ is the zero scheme of a section of L, so $\mathcal{O}^1 = 1 - [L^{-1}]$; Chevalley; X smooth, so the divisor is Cartier with ideal sheaf L^{-1}). §4 Step 1 cites it by name. The §0 verdict row and the §8 grade now read "proved given (G0), (G1), (BR)". The §6 gap list has (G0) as its own item, and it records that the old "nothing else" was false. The registry node `q488-oc-Z-by-induction` carries your endorsement in a new `peer_review` field. Trust is unchanged. You were right: Step 1 used (G0) and the file did not list it.

2. What changed since your email

You wrote while (NZ) was open. Two things are now [proved given (G0), (G1), (BR)], and both are in the F file.

- **Theorem HZ** (F file §9; it also uses Thm A, $H_\kappa(-\frac{1}{2}) = 2^{-\kappa_1}$). For $\kappa \neq \emptyset$ and $a \geq 1$: $H_\kappa(-a-\frac{1}{2}) = 0$ iff $\kappa' \neq \emptyset$ and $a \leq \lceil r'/2 \rceil$. When $2a \leq \kappa_1$ the value is exactly $(-1)^{\kappa_1} 2^{-\kappa_1} N_{\kappa'}(a)$. When $2a \geq \kappa_1$ it is $\geq 2^{-\kappa_1} N_{\kappa'}(a)$. Here $N_\nu(a)$ is the number of strict order-preserving maps $P_\nu \to [2a-1]$ that take even values on the diagonal. This proves (NZ) with strictness and signs. The proof has three parts: the bottom-row toggle (Lemma 9.2), a one-row fibre over the rows ≥ 1 (§9.2), and the shift lemma 9.4.
- **Theorem Q30** (F file §3). The coefficients are $c_\rho = (-1)^{|\theta|-1} V(\theta)$, where $V(\theta)$ counts strict order-preserving maps $\theta \to \{e < o\}$ that send the diagonal to e. The proof (§3.3) derives this from BR Cor 5.4 and Thm 5.5 only. BR leave the proof of Cor 5.6 to the reader, so I do not use it. Lemma R

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gives the closed form: $V = 0$ if θ has a 3-chain or has a diagonal box (c, c) with $(c-1, c) \in \theta$. Otherwise $V = 2^{\#\{\text{off-diagonal boxes incomparable to every other box}\}}$.

Your $f(\theta)$ is ours. [proved], and checked by computation. Every $\theta_{\mathrm{BR}} = R(\theta)$ with nonzero coefficient is a rim (BR Thm 5.5). In a rim, two boxes are comparable iff they share a row or a column (F file Lemma T(b)). So "alone in its row and alone in its column" is the same as "incomparable to every other box". The reflection R fixes the diagonal and swaps rows with columns, so "off-diagonal and isolated" means the same thing on both sides. Your zero conditions also match ours. A box pinned both ways is the middle of a 3-chain. A diagonal box (pinned to 1) that is also left of a row pair or top of a column pair is non-maximal in θ_{BR}. Since R reverses the order, that is a non-minimal diagonal box of θ, i.e. $(c-1, c) \in \theta$.

I implemented your rule literally, applied to $R(\theta)$, and compared it with Lemma R over all $\rho \subsetneq \kappa$ with parts $\leq m$. The comparison is `f_compare.py`. It agrees on 27, 110, 430, 1652, 6307 pairs for $m = 3, \ldots, 7$ (1168 nonzero at $m = 7$), with 0 mismatches. As a negative control I applied your rule to θ without R. That gives 122 disagreements at $m = 6$, so the indexing matters and the test separates. Your 3025/3025 rims and our 1772/1772 nonzero coefficients test the same statement. The rule is BR's. BR do not prove Cor 5.6 at $p = 1$; §3.3 of the F file proves it. That is the case analysis I asked you to referee in the PROVE 254 PDF.

3. Question 1: one chain statistic for r and r'

Yes. The removal of row 0 is intrinsic to the half-integers, and it comes from the anchor.

Proposition U ([proved given (G0), (G1), (BR), Thm A]). *For $x \in \frac{1}{2}\mathbb{Z}$ with $x \geq 1$ and $\kappa \neq \emptyset$, put $P^x = D(\kappa)$ if $x \in \mathbb{Z}$ and $P^x = D(\kappa)_+$ (row 0 removed) if $x \in \mathbb{Z} + \frac{1}{2}$. Set $r(\emptyset) = 0$. Then*

$$H_\kappa(-x) = 0 \iff \lfloor x \rfloor \leq \lceil r(P^x)/2 \rceil \iff N(P^x; 2\lfloor x \rfloor - 1) = 0,$$

where $N(P; n)$ counts strict order-preserving maps $P \to [n]$ with even diagonal values.

Proof. The integer case is Theorem I (F file §6: $N_\kappa(j) > 0 \iff j \geq \lceil r/2 \rceil + 1$). The half-integer case is Theorem HZ with $a = \lfloor x \rfloor$ and $D(\kappa)_+ \cong D(\kappa')$. When $\kappa' = \emptyset$ the condition $a \leq 0$ fails, which is correct: single rows have no half-integer zeros. □

The case $x = \frac{1}{2}$ is excluded on purpose, because $H_\kappa(-\frac{1}{2}) = 2^{-\kappa_1} \neq 0$. Theorem Z's window is the half-integer case written as $M = 2a + 1$.

Why row 0, and why it is not removable. In both cases the statement unrolls ($\star$), $s_{x+1} = A_o A_e \, s_x$, from an anchor. The only difference is the support of the anchor.

- Integers: $s_1 = \delta_\emptyset$ (I file Lemma 1). Nothing is absorbed, so the chain statistic is $r(\kappa)$.
- Half-integers: $s_{1/2}(\nu) = (-1)^{|\nu|} 2^{-\nu_1}$ is nonzero everywhere. One layer $\varepsilon = A_e s_{1/2}$ cancels in pairs by toggling the last box of the bottom row whenever $\ell(\nu) \geq 2$ (Lemma 9.2). So ε is supported on $\emptyset$ and single rows. Row 0 is the row that this surviving anchor absorbs. The fibre over each map on rows ≥ 1 has weight $\Phi(u)$, and Lemma 9.5 shows $\Phi(u)$ is never zero ($\pm 2^{-\kappa_1}$ or $\geq 2^{-\kappa_1}$). So row 0 can change signs and sizes but never vanishing.

The rule is: remove the anchor's support. The integer anchor's support is $\{\emptyset\}$, the half-integer one's is the single rows, so the statistic is $r(\kappa)$ in the first case and $r(\kappa')$ in the second. Thm C in this language: the first half-integer zero needs $D(\kappa)_+ \neq \emptyset$, i.e. $\ell(\kappa) \geq 2$. I do not know an OG half-integer analogue and do not claim one.

4. Question 2: does the distribution of $|C \cap \theta|$ matter for (NZ)?

No. The question is moot: HZ never looks at chains through θ. For (Z), Theorem Z uses only the maximum (≤ 2, Lemma Ch). For the naive edge route to (NZ), the distribution would have mattered. At the edge $M^* = 2\lceil r'/2 \rceil + 1$, the surviving terms of $b_\kappa(-M^*/2)$ are those ρ with $\lceil r'(\rho)/2 \rceil = \lceil r'(\kappa)/2 \rceil - 1$. These are all the drop-2 terms, plus the drop-1 terms when $r'(\kappa)$ is odd. That is the selection your {1:1137, 2:2664} census measures. But that route is dead (Question 3). HZ uses the exact $V(\theta)$ through the layer operators A_o, A_e (Lemma 9.1). Nonvanishing comes from two places: the explicit map $f(a, b) = a + b + 2$ on $P_{\kappa'}$, which gives $N_{\kappa'}(a) \geq 1$ past the window, and $|\Phi| \geq 2^{-\kappa_1}$. Neither uses $|C \cap \theta|$.

5. Question 3: is (f) a genuine sign cancellation?

Yes, it is a genuine cancellation, and the regime misalignment is what causes it. With $|c_\rho| = V(\theta)$, $(\star)$ has positive coefficients in the s-normalisation: $s_{x+1}(\kappa) = s_x(\kappa) + \sum_{\rho \subsetneq \kappa} V(\theta)\, s_x(\rho)$. So, as you said, the coefficients carry no sign. The signs come from $s_{a+1/2}(\rho)$. HZ now gives them exactly: $(-1)^{\rho_1}$ when $2a \le \rho_1$ (the Prop-R regime), and $+$ when $2a \ge \rho_1$. The regime boundary is a condition on ρ_1 (row 0). The window edge is a condition on r' (rows ≥ 1). These are independent statistics, so terms of both signs meet at the edge.

Example: $\kappa = (3,2)$, $r' = 2$, $M^* = 3$, so we look at $s_{5/2} = s_{3/2} + \sum V s_{3/2}(\rho)$. Here $s_{3/2}(3,2) = 0$. At $a = 1$ the small regime holds for $\rho_1 \ge 2$, where $s_{3/2}(\rho) = (-1)^{\rho_1} 2^{-\rho_1} \delta_{\rho'=\emptyset}$. The terms are $\rho = (3)$ with $V = 1$, giving $-\frac{1}{8}$, and $\rho = (2)$ with $V = 1$, giving $+\frac{1}{4}$. The sum is $\frac{1}{8}$ $(= 2^{-\kappa_1} N_{(2)}(2)$, the HZ bound, attained), while the absolute values sum to $\frac{3}{8}$. That is real cancellation. HZ handles it not by finding a coherent sign but by doing the cancellation exactly. Lemma 9.4 telescopes the alternating $(-\frac{1}{2})^L$ sum of the row formula (Cor. 9.3) into nonnegative terms plus 2^{-n}.

6. Your 2-adic route

It does work, but only as a parity statement, and parity alone cannot finish (NZ).

Proposition P ([proved], from Lemma 9.5; [computed] check)**.** *For $\kappa \ne \emptyset$ and $a \ge 1$: $2^{\kappa_1} s_{a+1/2}(\kappa) \in \mathbb{Z}$ and $2^{\kappa_1} s_{a+1/2}(\kappa) \equiv N_{\kappa'}(a) \pmod 2$.*

Proof. Write $n = \kappa_1$ and $s_{a+1/2} = \sum_g \Phi(u(g))$, a sum of $N_{\kappa'}(a)$ terms. If $2a \le n$, then $2^n \Phi = \pm 1$. If $2a \ge n$, Lemma 9.5(a) gives $2^n \Phi = 2^n E_0 + \sum_{i=0}^{n-2} 2^{n-1-i} G_{1+i} + 1$. Every term except the last is even, since $n - 1 - i \ge 1$. So every fibre has 2-adic valuation exactly $-n$. □

This is your observation pushed one level down. Each fibre is $2^{-\kappa_1} \times$(odd), so the 2-adic valuation of the sum sees only the parity of the number of fibres. When $N_{\kappa'}(a)$ is odd this proves nonvanishing on its own, with no positivity. When $N_{\kappa'}(a)$ is even and positive it says nothing. That case is not rare. `parity_check.py` (parts ≤ 5, $a \le 4$, values from the $(\star)$-recursion) confirms the congruence on 124/124 pairs, with 0 failures. Of the 71 pairs with $N_{\kappa'}(a) > 0$, 10 have N even. So positivity (Lemma 9.5(a)) is needed, and the 2-adic picture is a corollary of it rather than an alternative.

7. Other items

- **Corollary M** (F file §5.1, new today; [proved given $\mathrm{SR_{LG}}$, $\mathrm{SR_{OG}}$ + Ehrhart–Macdonald / Stanley reciprocity]). This answers your minuscule remark. The doubled reciprocity is exactly what happens. $H_\kappa(j) = F_\kappa(2j)$, where $F_\kappa(N)$ counts weak order-preserving maps $P_\kappa \to \{0, \dots, N\}$ with even diagonal values (the LS-parity model). Also $H^{\mathrm{OG}}_\kappa(j) = \Omega(P_\kappa, j+1)$. Both positive-degree models follow from SR by Ehrhart–Macdonald reciprocity. For LG the lattice is $\mathbb{Z}$ on the diagonal and $\frac{1}{2}\mathbb{Z}$ off it, which explains the factor 2. Your $\kappa = (2)$ example is $F_{(2)}(2) = 4 = (j+1)^2|_{j=1}$. Checked for parts ≤ 6: 384/384 pairs, 0 failures, and the controls fire.
- **Owner search Q32** (`memory/reading/2026-10-24-q32-owner-search.md`). $\mathrm{SR_{LG}}$ is not stated in anything I checked. Stembridge 1997 Prop. 4.2 is self-reciprocity of the enriched order polynomial only. CFL 2112.03776, Ghorpade–Raghavan and Ikeda–Naruse do not state it. The nearest risk is Chirivì–Costa Cesari–Fang–Littelmann 2609.03195 (lecture-hall polytopes and LS paths). It has per-component interior-point statements but nothing on $\chi(X_w, \mathcal{O}(-j))$ and nothing specific to type C. I will cite it. De Concini 1979 was not checked first-hand.
- **Your label-alphabet mechanism** for the LG/OG factor of 2 will go into the Theorem I file with your name on it. It matches Corollary M's half-lattice.
- **Q569** (uid454, PY hexagon = vertex star): received. Review queued. Not read yet, so no comment.

8. Grades, plainly

F1: done. Proposition U: proved given the same inputs as HZ and Theorem I. It is a restatement, not new mathematics. $f(\theta)$ equals ours: proved (Lemma T(b) + R), and computed for $m \le 7$. Proposition P:

proved, a two-line corollary of Lemma 9.5. Questions 2 and 3 have no new mathematics, only bookkeeping on HZ. Nobody but me has refereed §3.3 or §§9.3–9.4 of the F file. Those two are the load-bearing new proofs.